EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

Practice Assignment – Quarter 1, 2026- 2027

Course Name: Coding Practice
Course Code: CSE 422
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LAB 05

GitHub Link: https://github.com/thaiphanthanhsang/CSE422_CodingPractice.git

1. Introduction

This project is a simple library management system implemented as a C# console application. It uses Microsoft SQL Server to store documents, users, and borrowing transactions. The main purpose of the laboratory is to demonstrate four object-oriented design patterns through a practical application.

2. Project Objectives

- Add and display library documents (books, magazines, and newspapers).
- Add and display users.
- Borrow available documents, calculate the loan fee, and reduce stock.
- Return borrowed documents, update the loan status, and restore stock.
- View loan records and display notifications to subscribed observers.
- Keep classes in separate C# files to make the application easier to understand.

3. Database Design

Table	Main columns	Purpose
Documents	Id, Title, DocumentType, Quantity	Stores document types and available stock.
Users	Id, Name	Stores registered library users.
Loans	Id, DocumentId, UserId, BorrowDate, ReturnDate, Days, Fee, IsReturned	Stores borrowing and returning transactions.

4. Singleton Parttern

Purpose

Singleton ensures that the application uses one instance of a database-management class. It centralizes the SQL Server connection string and provides a method to open SQL connections.

Implementation

File: Database/DatabaseConnection.cs

```
using Microsoft.Data.SqlClient;
namespace LibraryDesignPatterns.Database;

// Singleton: only one database manager object.
6 references
public sealed class DatabaseConnection
{
    1 reference
    private static readonly DatabaseConnection _instance = new();
    1 reference
    public static DatabaseConnection Instance => _instance;
    1 reference
    private DatabaseConnection() { }

    // Change .\SQLEXPRESS to your SQL Server instance name.
    1 reference
    private readonly string _connectionString =
        @"Server=(localdb)\MSSQLLocalDB;Database=LibraryDB;Trusted_Connection=True;Encrypt=True;TrustServerCertificate=True;";

    // SQL connections are short-lived and reused internally through pooling.
    7 references
    public SqlConnection OpenConnection()
    {
        var connection = new SqlConnection(_connectionString);
        connection.Open();
        return connection;
    }
}
```

The constructor is private, so other classes cannot instantiate DatabaseConnection directly. LibraryService accesses the shared instance through DatabaseConnection.Instance. Each operation opens a separate short-lived SqlConnection; the singleton is the manager, not one permanently open physical database connection.

5. Factory Method Pattern

Purpose

Factory Method delegates document creation to specialized creator classes, so the calling code does not need to construct Book, Magazine, or Newspaper directly.

Implementation

Files: FactoryMethod/IDocument.cs, DocumentCreator.cs, BookCreator.cs, MagazineCreator.cs, NewspaperCreator.cs, Book.cs, Magazine.cs, Newspaper.cs

```
namespace LibraryDesignPatterns.FactoryMethod;
8 references
public interface IDocument
{
    8 references
    string Title { get; }
    5 references
    string Type { get; }
}
```

```
namespace LibraryDesignPatterns.FactoryMethod;
Factory Method: subclasses decide which document to create.
4 references
public abstract class DocumentCreator
{
    4 references
    public abstract IDocument Create(string title);
}
```

```
namespace LibraryDesignPatterns.FactoryMethod;
2 references
public class Book : IDocument
{
    4 references
    public string Title { get; }
    3 references
    public string Type => "book";
    1 reference
    public Book(string title) => Title = title;
}
```

```
namespace LibraryDesignPatterns.FactoryMethod;
1 reference
public class BookCreator : DocumentCreator
{
    2 references
    public override IDocument Create(string title) => new Book(title);
}
```

The DocumentCreator abstraction defines the factory method. Each concrete creator implements it to return the matching document type. The service selects a creator based on input and calls Create(title).

6. Observer Pattern

Purpose

Observer lets subscribers receive a message when library activity occurs, without hard-coding a notification for each user inside the borrowing logic.

Implementation

Files: Observer/ISubscriber.cs, UserSubscriber.cs, LibraryNotifier.cs

```
namespace LibraryDesignPatterns.Observer;
3 references
public interface ISubscriber
{
    2 references
    void Update(string message);
}
```

```
namespace LibraryDesignPatterns.Observer;
2 references
public class LibraryNotifier
{
    2 references
    private readonly List<ISubscriber> _users = new();
    1 reference
    public void Subscribe(ISubscriber user) => _users.Add(user);
    3 references
    public void Notify(string message)
    {
        foreach (var user in _users) user.Update(message);
    }
}
```

```
namespace LibraryDesignPatterns.Observer;
2 references
public class UserSubscriber : ISubscriber
{
    2 references
    public string Name { get; }
    1 reference
    public UserSubscriber(string name) => Name = name;
    2 references
    public void Update(string message) => Console.WriteLine($"[Notification to {Name}] {message}");
}
```

The notifier keeps an in-memory collection of subscribers. Each UserSubscriber implements the update operation and prints the message in the console. In this simple version, subscriptions last only for the current program session and notifications are sent to all registered subscribers; document-specific subscription filtering is not implemented.

7. Strategy Pattern

Purpose

Strategy separates the borrowing-fee formula from the borrowing workflow. Different document types can use different rates without changing the Loan calculator

Implementation

Files: Strategy/IFeeStrategy.cs, BookFeeStrategy.cs, MagazineFeeStrategy.cs, NewspaperFeeStrategy.cs, Loan.cs

```
namespace LibraryDesignPatterns.Strategy;
1 reference
public class BookFeeStrategy : IFeeStrategy
{
    2 references
    public decimal Calculate(int days) => days * 2000m;
}
```

```
namespace LibraryDesignPatterns.Strategy;
6 references
public interface IFeeStrategy
{
    4 references
    decimal Calculate(int days);
}
```

```
namespace LibraryDesignPatterns.Strategy;
2 references
public class Loan
{
    2 references
    private readonly IFeeStrategy _strategy;
    1 reference
    public Loan(IFeeStrategy strategy) => _strategy = strategy;
    1 reference
    public decimal GetFee(int days) => _strategy.Calculate(days);
}
```

```
namespace LibraryDesignPatterns.Strategy;
1 reference
public class MagazineFeeStrategy : IFeeStrategy
{
    2 references
    public decimal Calculate(int days) => days * 1500m;
}
```

For example, borrowing a book for five days costs 5 × 2,000 = 10,000 VND. The strategy object is chosen from DocumentType and then passed to Loan.

8. Demo Project

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 1
Type (book/magazine/newspaper): book
Title: A
Quantity: 4
Document added.
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 2
1 | A | book | Available: 4
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 3
Name: Sang
User added.

=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 3
Name: Linh
User added.
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 4
1 | Sang
2 | Linh
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 5
Document ID: 1
User ID: 1
Days: 3
book A borrowed by Sang. Fee: 6,000 VND
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 6
Loan ID: 1
Book A returned by Sang
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 7
Loan 1 | Sang | A | 3 days | 6,000 VND | Returned
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 8
Your name: Sang
Subscribed for this session.

=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 8
Your name: Linh
Subscribed for this session.
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 5
Document ID: 1
User ID: 1
Days: 3
book A borrowed by Sang. Fee: 6,000 VND
[Notification to Sang] book A borrowed by Sang
[Notification to Linh] book A borrowed by Sang
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 7
Loan 1 | Sang | A | 3 days | 6,000 VND | Returned
Loan 2 | Sang | A | 3 days | 6,000 VND | Borrowed
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 6
Loan ID: 2
Book A returned by Sang
[Notification to Sang] Book A returned by Sang
[Notification to Linh] Book A returned by Sang
```

```
=== LIBRARY MANAGEMENT ===
1. Add document
2. View documents
3. Add user
4. View users
5. Borrow document
6. Return document
7. View loans
8. Subscribe notification
0. Exit
Choose: 7
Loan 1 | Sang | A | 3 days | 6,000 VND | Returned
Loan 2 | Sang | A | 3 days | 6,000 VND | Returned
```